DETECTION OF THE MALICIOUS WEB PAGES USING MACHINE LEARNING.

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BACHELOR OF ENGINEERING

IN

COMPUTER ENGINEERING

SCHOOL OF INFORMATION TECHNOLOGY
MAE FAH LUANG UNIVERSITY
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A SENIOR PROJECT SUBMITTED TO

MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF

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THIS SENIOR PROJECT HAS BEEN APPROVED

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Abstract

Because there are so many dangerous websites these days. Those dangerous websites tend to cause trouble for many people. Whether to deceive money transfer Or tricked into filling out transaction information So, how do we know which websites are dangerous? We, therefore, apply machine learning knowledge to audit websites from around the world. The random forest algorithm uses python SKlearn library to calculate the probability of malicious website. And using a dataset consisting of many website domains to calculate the properties of the Domain to determine what type of content and activity is likely to be found on malicious websites. Our results demonstrate the effectiveness of our concept. This approach can be further researched. By the site administrator or system administrator to identify potentially dangerous websites to protect users from online threats.

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CHAPTER 1
INTRODUCTION

1.1 Background and Rational

A website is a web page that is created to present information via the Internet with many web pages linked together. To be able to go to various websites, users can choose to see interesting information, just search, and choose what they want. We can't know if the website is safe.
unsafe website Just by clicking on that website, we have the opportunity to be hacked. Our data can cause damage to a certain extent. Therefore, we have made a website inspection system to separate dangerous websites for users, allowing users to know whether the website we are using is safe or not.

1.2 Objective

1.2.1 To identify bad websites.
1.2.2 To prevent being hacked various information from the website.
1.2.3 To be useful in various applications such as cybersecurity, web filtering and fraud detection.

1.3 Scope

1.3.1 Can detect malicious web pages using machine learning.
1.3.2 Can predict the percentage score of bad and good words that websites.

1.4 Plan

Table 1.1 Working plan

	Computer Engineering Pre-Project					Computer Engineering Project				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and										

requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										
Senior Project 2 exam										

Table 1. Working plan

1.5 Expected Result

1.6.1 plan.system can identify the words of websites.

1.6 Place and Equipment

1.7.1 Place(Locate the working place at MFU)

1.7.2 Hardware (Laptop or Computer)

1.7.3 Software(Docker)

Chapter2
LITERATURE REVIEW

2.1 Theory

Random forest is a type of machine learning model developed from Decision Tree. The difference is that Random Forest increases the number of trees into many trees, resulting in higher work efficiency. More accurately, the Random Forest model is a model that has become very popular in using Machine Learning.[1]

The advantage of Random Forests is that they can be used with both structured (columns and tables) and unstructured (images, text) data, and can prevent overfitting of the dataset from which they were created. Random forests are very useful for detecting bad words on a website because the theory works by randomly selecting features and data from all available datasets and using them to predict the results. When complete, use the voting results to make the decision. If it is predicted that there are more bad words than good words, then this means that the website is not safe due to the presence of bad words[6][8].

2.2 Related Research

(Dipankar Kumar Mondal and Bikash Chandra Singh, 2021) Malicious websites are the threats web security brings to users. An efficient and effective model for detecting malicious URLs has been developed. To provide notifications when a user accesses using machine learning methods, however, no technique is completely detectable. Because these are susceptible to false positives and negatives decisions to classify URLs in malicious and non-malicious groups. Therefore, the method of multiple classifiers is used to predict the probabilities of URL classes, and then the criteria are applied. to filter decisions on multiple classifiers. It then applies decisions along with probabilities to show the safety of that URL[3].

2.3 Related Work

Related work of Detection of malicious web pages using machine learning. The relevant departments can examine various classification attributes such as domain, terminology, and content.

The study uses the random forests algorithm to detect certain types of dangerous web pages, such as phishing, malware, etc. Therefore, web page classification and security such as anti-virus protection are highly effective. In this section, this report can be further developed. and evaluating the potential of random forests for practical use.

The latest development of the random forests algorithm is research on the scalability, robustness, and interpretability of the random forests -related work classifier. The potential of these developments can be explored to catch malicious web pages

2.4 Related Technology

2.4.1 Docker

Docker is a software platform built from out-of-the-box testing of production applications. Docker standardizes required target software known as containers.

Figure 2. 1 Logo docker image

2.4.2 Microsoft Excel

Microsoft Excel is a spreadsheet program that has the ability to create tables, calculations, analysis, and reporting in the form of tables and graphs.

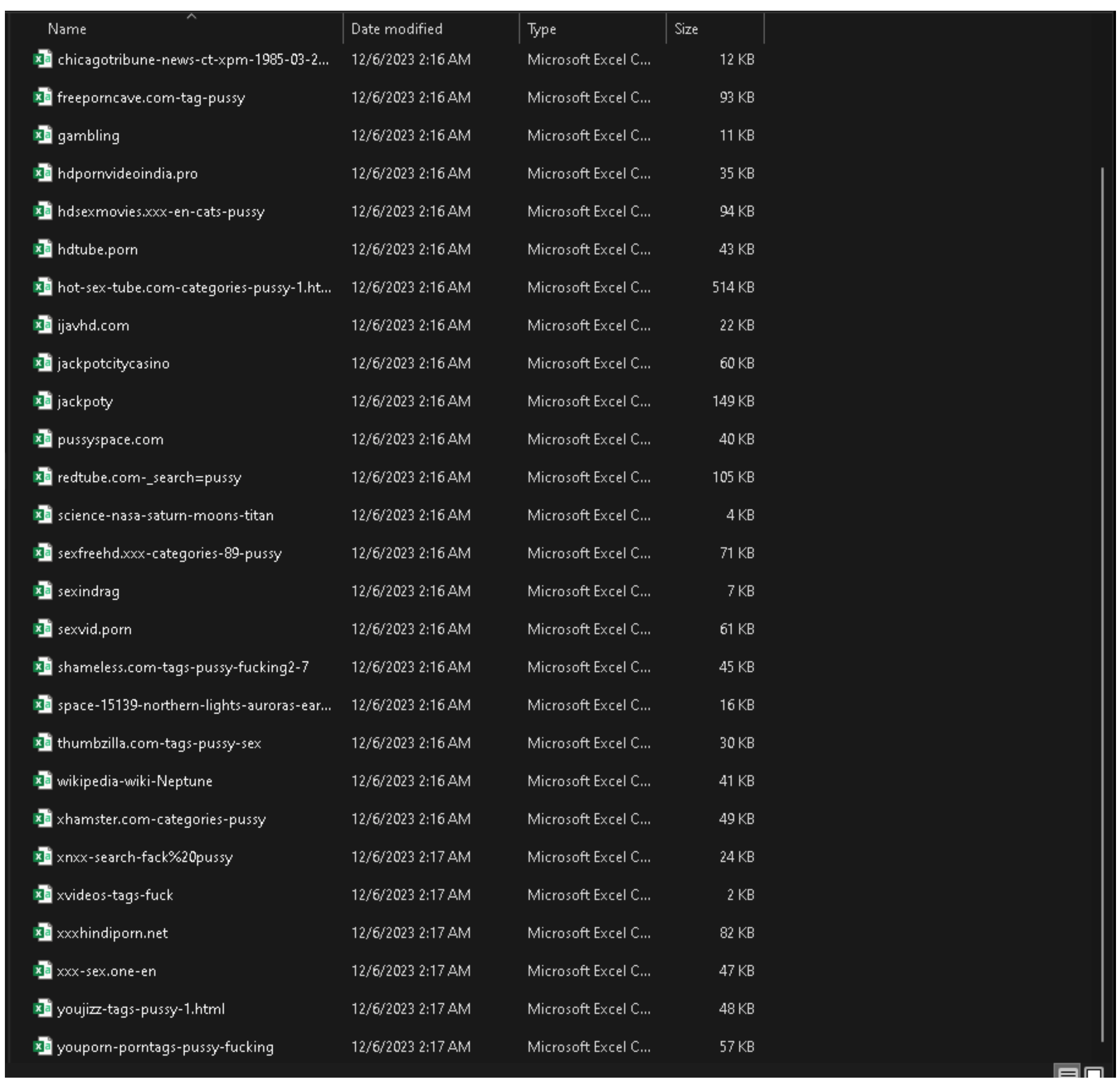
Figure 2. 2 Logo Excel image

CHAPTER 3
METHODOLOGY

3.1 Extract website to file CSV

3.1.1 Extract website to file CSV

We extract words from websites by BeautifulSoup this library helps us pulling data out of HTML and XML files So, we extract data to 1 word for 1 cell and only 1 column in horizontal.



Name	Date modified	Type	Size
chicagotribune-news-ct-xpm-1985-03-2...	12/6/2023 2:16 AM	Microsoft Excel C...	12 KB
freeporncave.com-tag-pussy	12/6/2023 2:16 AM	Microsoft Excel C...	93 KB
gambling	12/6/2023 2:16 AM	Microsoft Excel C...	11 KB
hdpornvideoindia.pro	12/6/2023 2:16 AM	Microsoft Excel C...	35 KB
hdsexmovies.xxx-en-cats-pussy	12/6/2023 2:16 AM	Microsoft Excel C...	94 KB
hdtube.porn	12/6/2023 2:16 AM	Microsoft Excel C...	43 KB
hot-sex-tube.com-categories-pussy-1.ht...	12/6/2023 2:16 AM	Microsoft Excel C...	514 KB
ijavhd.com	12/6/2023 2:16 AM	Microsoft Excel C...	22 KB
jackpotcitycasino	12/6/2023 2:16 AM	Microsoft Excel C...	60 KB
jackpoty	12/6/2023 2:16 AM	Microsoft Excel C...	149 KB
pussyspace.com	12/6/2023 2:16 AM	Microsoft Excel C...	40 KB
redtube.com-_search=pussy	12/6/2023 2:16 AM	Microsoft Excel C...	105 KB
science-nasa-saturn-moons-titan	12/6/2023 2:16 AM	Microsoft Excel C...	4 KB
sexfreehd.xxx-categories-89-pussy	12/6/2023 2:16 AM	Microsoft Excel C...	71 KB
sexindrag	12/6/2023 2:16 AM	Microsoft Excel C...	7 KB
sexvid.porn	12/6/2023 2:16 AM	Microsoft Excel C...	61 KB
shameless.com-tags-pussy-fucking2-7	12/6/2023 2:16 AM	Microsoft Excel C...	45 KB
space-15139-northern-lights-auroras-ear...	12/6/2023 2:16 AM	Microsoft Excel C...	16 KB
thumbzilla.com-tags-pussy-sex	12/6/2023 2:16 AM	Microsoft Excel C...	30 KB
wikipedia-wiki-Neptune	12/6/2023 2:16 AM	Microsoft Excel C...	41 KB
xhamster.com-categories-pussy	12/6/2023 2:16 AM	Microsoft Excel C...	49 KB
xnxx-search-fack%20pussy	12/6/2023 2:17 AM	Microsoft Excel C...	24 KB
xvideos-tags-fuck	12/6/2023 2:17 AM	Microsoft Excel C...	2 KB
xxxhindiporn.net	12/6/2023 2:17 AM	Microsoft Excel C...	82 KB
xxx-sex.one-en	12/6/2023 2:17 AM	Microsoft Excel C...	47 KB
youjizz-tags-pussy-1.html	12/6/2023 2:17 AM	Microsoft Excel C...	48 KB
youporn-porntags-pussy-fucking	12/6/2023 2:17 AM	Microsoft Excel C...	57 KB

Figure 3. 1 Example csv files of websites.

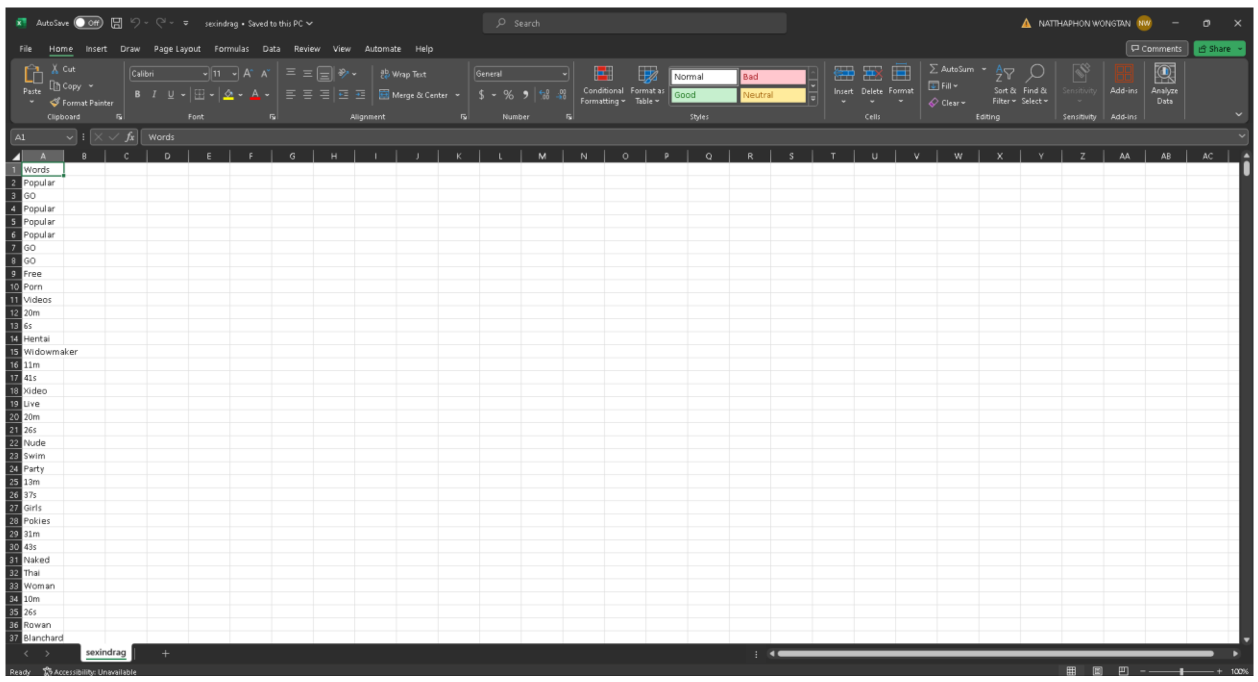


Figure 3. 2 Example words extract from websites.

3.1.2 Bad and Good words dataset

We download dataset of bad and good words from Kaggle. This data has a lot of datasets that we want. We give number 1 for bad words and number 0 for good words because machine learning can read only binary numbers then we shuffle bad and good words. We make this dataset for doing algorithm of machine learning.

```
import pandas as pd

# Load the dataset from the CSV file
file_path = 'data/words.csv'
data = pd.read_csv(file_path)

# Shuffle the dataset
shuffled_data = data.sample(frac=1, random_state=42)  # frac=1 shuffles the entire dataset

# Save the shuffled dataset back to a CSV file if needed
shuffled_data.to_csv('shuffled_dataset.csv', index=False)
```

Figure 3. 3 Shuffle function.

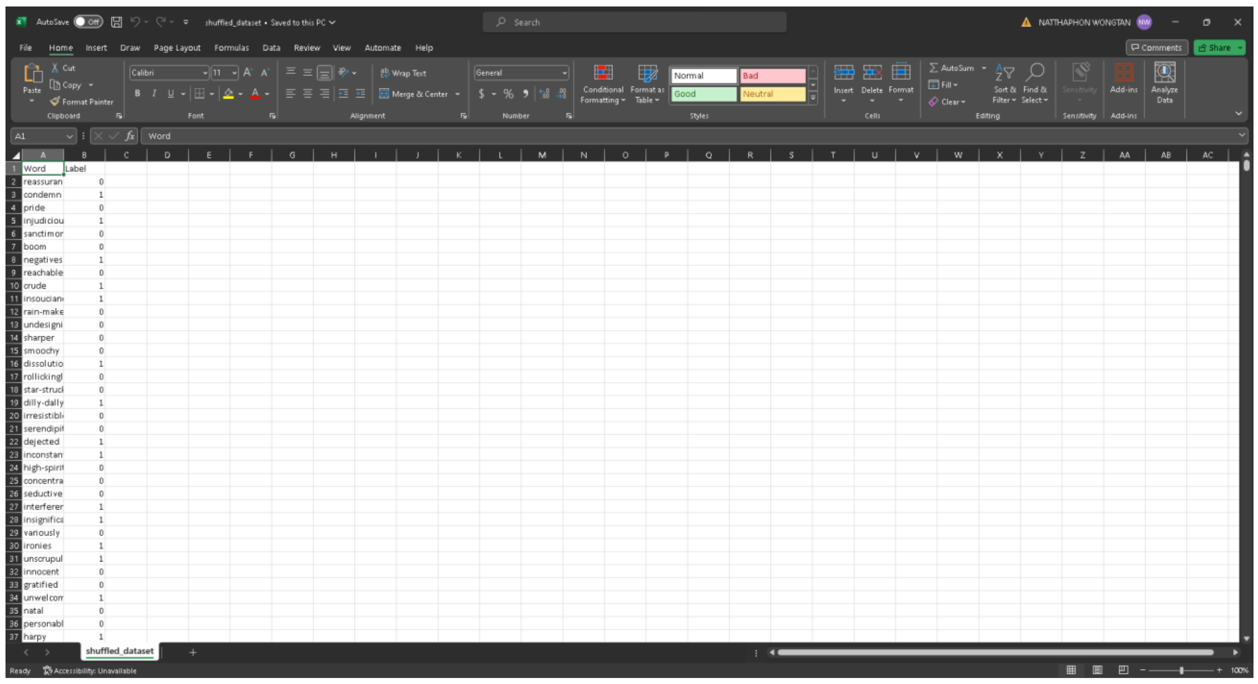


Figure 3. 4 Shuffle words bad and good.

3.2 Machine learning algorithm

We have dataset of words then we will use five algorithms according to the chatbot's advice and from what we have tried to study we should use these algorithms.

3.2.1 Naive Bayes Classifier

This model works well for text classification tasks and is relatively simple and fast. It's based on Bayes' theorem and assumes independence among features[5].

3.2.2 Support Vector Machines (SVM)

SVMs are good for binary classification tasks like positive/negative sentiment analysis. They try to find the best hyperplane that separates the data into different classes[2].

3.2.3 Logistic Regression

Logistic Regression tends to perform well with medium-sized datasets. It's relatively fast and can handle thousands of records efficiently[7].

3.2.4 Neural Networks (Deep Learning)

With enough computational power and optimized implementations (such as using frameworks like TensorFlow or PyTorch), neural networks can handle large datasets effectively and may provide high accuracy. However, they might be computationally intensive[4].

3.2.5 Random Forest

With enough computational power and optimized implementations (such as using frameworks like TensorFlow or PyTorch), neural networks can handle large datasets effectively and may provide high accuracy. However, they might be computationally intensive.

3.3 Result

The algorithm we selected is Random Forest because the recall and precision is more than other algorithms. Then when we have model, we select the dataset of website to watch the bad and good words So, bad words are 1 and good is 0.

```
[46]: import pandas as pd
      data = pd.read_csv('extracted_words_BAD.csv')

[47]: # Text vectorization
      from sklearn.feature_extraction.text import CountVectorizer
      #vectorizer = CountVectorizer()
      vectorizer = joblib.load('vectorizer_model.joblib')
      X_train_vec = vectorizer.fit_transform(data['Words'])

[48]: import joblib
      model = joblib.load('Randomforest_model.joblib')
      model

[48]:       RandomForestClassifier
      RandomForestClassifier(max_depth=30)

[49]: import pandas as pd
      from sklearn.feature_extraction.text import CountVectorizer
      import joblib

      # Load CSV data
      data = pd.read_csv('./exemple-web-goodorbad/hdpornvideoindia.pro.csv')

      # Load the trained vectorizer
      vectorizer = joblib.load('vectorizer_model.joblib')

      # Text vectorization
      X_new_vec = vectorizer.transform(data['Words'])

      # Load the trained Random Forest model
      model = joblib.load('Randomforest_model.joblib')

      # Make predictions on the new data
      y_pred = model.predict(X_new_vec)

[53]: # Calculate percentages
      percentages = dataf.value_counts(normalize=True) * 100

      # Print results
      print("Percentages:")
      print(percentages)

      Percentages:
      1    83.330263
      0    16.669737
      dtype: float64
```

Figure 3. 5 Result of detected.

CHAPTER 4
EXPERIMENT AND RESULTS

In this Project of detection of malicious web pages using machine learning we watch the result from percent of words bad and then we will identify websites that are bad or good .

4.1 Performance of Machine Learning Model

Random forest is the best for this project because precision is the best score. Then if we have a model we can watch the result.

	precision	recall	f1-score	support
0	0.89	0.08	0.15	926
1	0.53	0.99	0.69	958
accuracy			0.54	1884
macro avg	0.71	0.54	0.42	1884
weighted avg	0.71	0.54	0.42	1884

Figure 4. 1 Predict of Random Forest.

```
# Calculate percentages
percentages = dataf.value_counts(normalize=True) * 100

# Print results
print("Percentages:")
print(percentages)
```

```
Percentages:
1    83.330263
0    16.669737
dtype: float64
```

Figure 4. 2 Example result from websites.

4.2 Comparison with Other Methods

As for other methods, the effect of precision will be less than in the method of random forests

Deep learning

	precision	recall	f1-score	support
0	0.48	0.95	0.64	926
1	0.25	0.01	0.03	958
accuracy			0.48	1884
macro avg	0.37	0.48	0.33	1884
weighted avg	0.36	0.48	0.33	1884

Figure 4. 3 Predict of Deep learning.

Logistics regression

	precision	recall	f1-score	support
0	0.80	0.13	0.23	926
1	0.54	0.97	0.69	958
accuracy			0.56	1884
macro avg	0.67	0.55	0.46	1884
weighted avg	0.67	0.56	0.46	1884

Figure 4. 4 Predict of Logistic regression.

Naive bayes

	precision	recall	f1-score	support
0	0.49	0.98	0.65	926
1	0.35	0.01	0.02	958
accuracy			0.49	1884
macro avg	0.42	0.49	0.34	1884
weighted avg	0.42	0.49	0.33	1884

Figure 4. 5 Predict of Naïve bayes.

SVM (support vector machine)

	precision	recall	f1-score	support
0	0.78	0.12	0.20	926
1	0.53	0.97	0.69	958
accuracy			0.55	1884
macro avg	0.66	0.54	0.45	1884
weighted avg	0.66	0.55	0.45	1884

Figure 4. 6 Predict of SVM (support vector machine).

4.3 Challenges and Limitations

The limitation of this project is that we can't identify or limit this word as a bad word with 100% accuracy, because it requires more effective translation machines, and because some words are not vulgar, they can be dangerous, so we must consider the words in sentence form.

4.4 Future Work

In future projects, we will add a sentence detection section because if words are used alone, it will not be detected. Therefore, it will be treated as a sentence to improve performance and be able to detect bad images. Some websites may not contain harmful or bad words, but may contain inappropriate images.

CHAPTER 5
CONCLUSION AND DISCUSSION

5.1 Conclusion

Throughout the duration of the project Malicious web page detection using machine learning. We found out what algorithms we will use to detect malicious web pages. We have seen from several studies that random forests are highly versatile and highly accurate. We, therefore, chose to use random forests for this research. Our workflow is to take the ML to train each website and let the AI analyze the words or content of each website and predict the percentage accuracy to easily determine if the website is malicious or not, To prevent malicious malicious websites Our research can be applied to security systems such as attack prevention, etc. in system development. We encountered many system problems. But we have a reasonable solution. This makes our work complete and is the starting point for further development in this line of work.

5.2 Discussion

Working on this project, we encountered several problems in the project. The first problem was to verify that the web was safe. We don't just have to check for words on the index page. But still need to detect other pages. The second problem is that we can't filter out 100% unsafe words because they are sensitive and difficult for AI to decide. Therefore, human judgment must be used as another decision.

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